RESEARCH REPORT

Customer Segmentation and Churn Pattern Analytics in European Banking

A data-driven study of customer retention and account balance exposure

Undergraduate research project | BBA Financial Markets

Dataset: European_Bank.csv | 10,000 customer records | Recorded year: 2025

Prepared: November 2026

This report analyzes a provided banking customer dataset. It does not claim to represent the European banking population or a verified financial institution. Currency is not specified in the source file.

Abstract

This study examines customer churn across 10,000 records from a European banking customer dataset covering France, Germany and Spain. The analysis combines data-quality checks, descriptive segmentation and account balance exposure measures. Customers are grouped by geography, age, credit score, tenure, account balance, product count and activity status. A separate top-20% balance group is used as a practical indicator of high-balance customers.

The overall recorded churn rate is **20.37%** (2,037 customers). Germany has the highest country-level rate at **32.44%**. Customers aged 46–60 show **51.12%** churn, while the Germany × age 46–60 segment reaches **67.33%**. Inactive customers have a **26.85%** churn rate versus **14.27%** among active customers. Two-product customers show relatively low churn (7.58%), while the small four-product group has 100% recorded churn; this unusual pattern should not be interpreted as causal.

Recorded balances associated with churned customers total **185.59 million dataset currency units**. The top 20% of customers by balance account for **38.62%** of those balances. The report recommends targeted investigation and careful retention testing, particularly for middle-aged customers and high-exposure segments. Results are observational and cannot establish the causes of churn or actual deposit losses.

Keywords

Customer churn; banking analytics; customer segmentation; exploratory data analysis; financial exposure; customer retention.

1. Introduction

Customer retention matters in retail banking because customers use accounts, cards and other financial services over long periods. Losing a customer can interrupt an established relationship and create additional acquisition and servicing costs. However, customer exits are not equally distributed across demographic and behavioral groups. An overall churn percentage can therefore hide important differences.

This project uses a structured customer dataset to explore where exits are concentrated. The focus is on identifying patterns that bank managers could investigate, not on claiming that the dataset describes all European banks. The three recorded geographies are France, Germany and Spain, which allow country-level comparisons within the sample.

1.1 Why segmentation matters

A single retention campaign is unlikely to fit every customer. A new customer with a small balance may have different needs from a long-standing customer holding multiple products. Segment-level analysis helps identify groups with high churn rates and groups contributing the greatest number of exits or associated balances. These are different management questions and should be reported separately.

1.2 Research scope

The work covers exploratory analysis and a companion interactive Streamlit dashboard. It does not include interviews, transaction histories, product pricing, customer complaints, market-share estimates or causal experiments. Those sources would be needed before recommending large operational changes.

2. Research Objectives and Questions

The central objective is to describe how recorded customer churn varies across customer groups and to identify where churn is associated with relatively large account balances.

Research questions

• How does the churn rate differ across France, Germany and Spain?

• Which age, tenure and credit score groups show higher or lower churn?

• How do customer activity and the number of products held relate to churn?

• How much recorded account balance is associated with exited customers?

• Which high-balance segments deserve closer management review?

Analytical measures

Churn rate = exited customers / total customers in the segment × 100. **Churn contribution** = exited customers in a segment / all exited customers × 100. **Relative risk index** = segment churn rate / overall churn rate. **Churn-associated balance** = sum of recorded balances for customers marked Exited = 1. None of these measures, on its own, establishes causation or actual monetary loss.

3. Data Source and Quality

The analysis uses the supplied file **European_Bank.csv**, containing 10,000 rows and 14 original fields. The file includes customer identifiers, geography, demographic information, account characteristics, estimated salary and an Exited indicator. All records have Year = 2025. The data source does not provide documentation establishing whether the records are actual bank customers, a teaching sample or simulated observations.

Field category	Original variables
Identification	CustomerId, Surname, Year
Demographic	Geography, Gender, Age
Account profile	CreditScore, Tenure, Balance, NumOfProducts
Engagement	HasCrCard, IsActiveMember
Other	EstimatedSalary, Exited

3.1 Quality checks

The dataset contains no missing values, no duplicate full rows and no duplicate customer IDs. Age ranges from 18 to 92, credit score from 350 to 850, and balances and estimated salaries are nonnegative. Binary fields use valid 0/1 values. These checks support the mechanical consistency of the file but do not establish external accuracy.

3.2 Interpretation limits

The meaning of the Exited flag is not documented beyond its binary coding. There is no recorded exit date, bank name, account type or currency. In particular, the balance recorded for an exited customer cannot be assumed to equal funds withdrawn or revenue lost.

4. Segmentation Methodology

Segments were defined before the main dashboard work so the same rules could be applied consistently in the scripts and report. Age groups were set to under 30, 30–45, 46–60 and above 60. Tenure was grouped as new (0–2 years), mid-term (3–6) and long-term (7–10). Credit scores were grouped as low (below 580), medium (580–699) and high (700 or above).

Balances were classified as zero, low positive and high positive. The positive-balance median was **119,839.69**; balances above that threshold were labeled high. For a separate high-value analysis, customers at or above the 80th percentile of all balances (**133,710.36**) were selected. This top-20% classification is a balance proxy, not a profitability or wealth measure.

Dimension	Operational definition
Age	Under 30; 30–45; 46–60; Above 60
Tenure	0–2; 3–6; 7–10 years
Credit score	Low <580; Medium 580–699; High ≥700
Balance	Zero = 0; Low = positive ≤119,839.69; High >119,839.69
High-value proxy	Top 20% of recorded balances, threshold 133,710.36
Churn	Exited = 1; retained = Exited = 0

4.1 Analytical approach

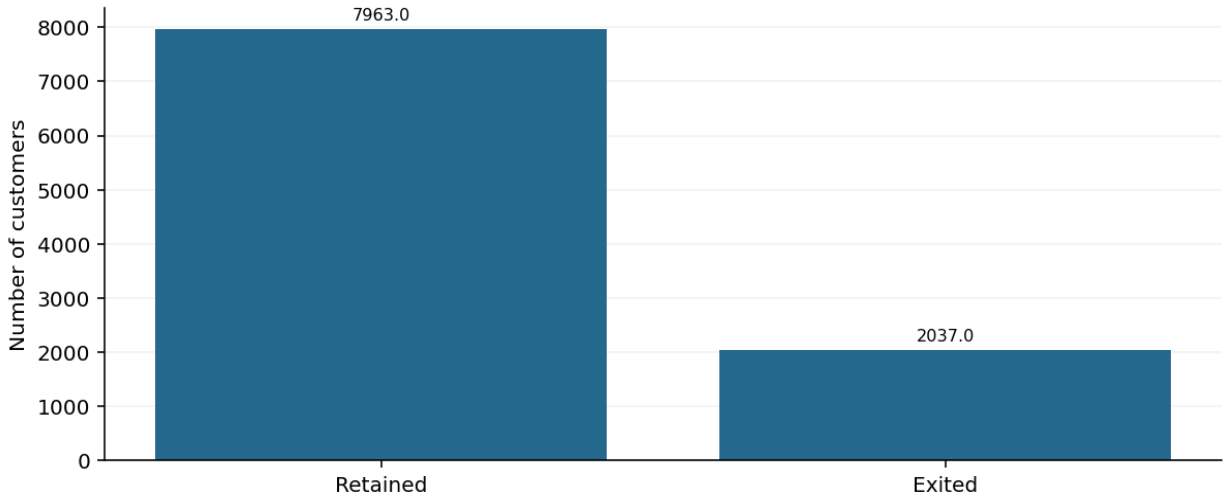
The workflow consisted of loading and validating the raw CSV, generating segmentation labels, computing grouped counts and rates, cross-checking totals, and exporting summary files. The dashboard uses the same record-level logic so interactive filters can reproduce the reported totals.

5. Overall Churn and Customer Profile

Of the 10,000 customers, 2,037 are marked as exited and 7,963 as retained. The overall churn rate is **20.37%**, with retention at **79.63%**.

Metric	Retained	Exited
Customers	7,963	2,037
Average age	37.41	44.84
Average credit score	651.85	645.35
Average balance	72,745.30	91,108.54
Average tenure	5.03	4.93

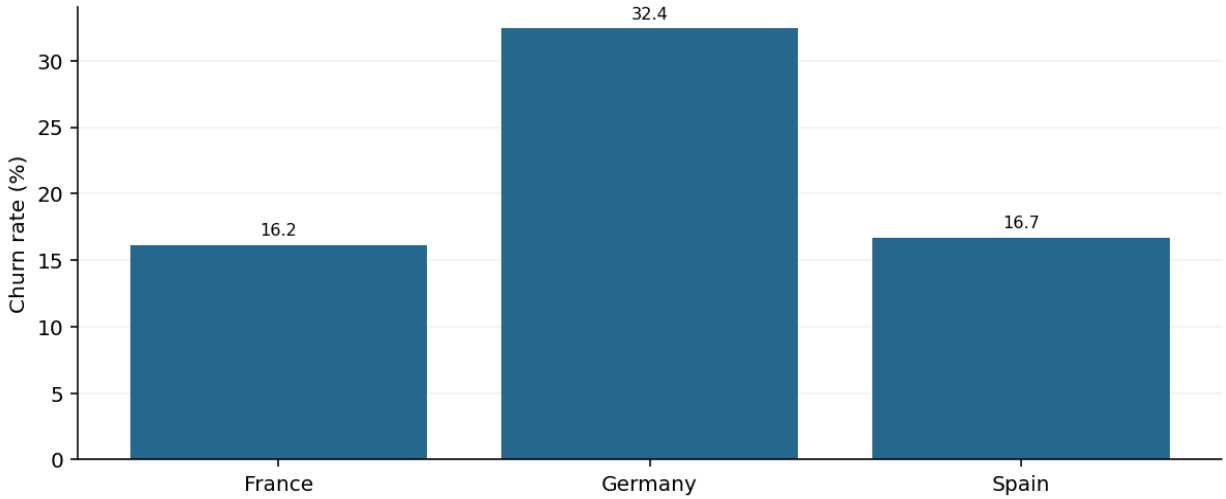
Exited customers are older on average (44.84 years) than retained customers (37.41 years). Their average recorded balance is also higher. These differences are descriptive and may reflect the different country and age composition of the groups.



Source: calculations from European_Bank.csv (10,000 records, Year = 2025).

6. Geographic Churn

Country	Customers	Exited	Churn rate
France	5,014.0	810.0	16.15%
Germany	2,509.0	814.0	32.44%
Spain	2,477.0	413.0	16.67%



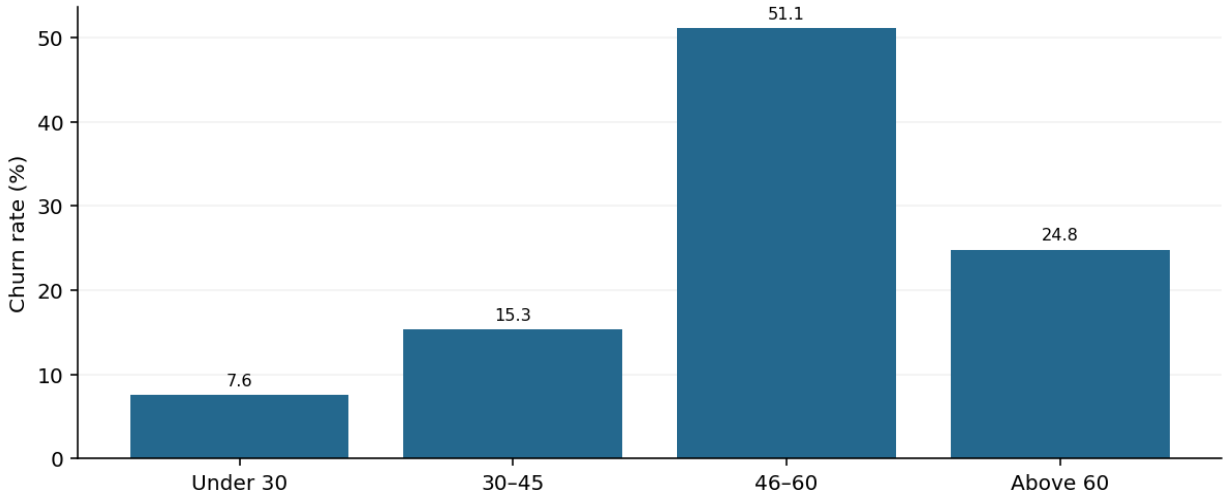
Source: calculations from European_Bank.csv (10,000 records, Year = 2025).

Germany records the highest churn rate, at 32.44%, compared with 16.15% in France and 16.67% in Spain. Germany represents roughly one-quarter of all customers but accounts for approximately 40% of exits. France contributes a similar number of exits because its customer population is much larger.

This difference is worth investigation, but the dataset does not reveal whether it is related to service experience, market conditions, customer selection or how the records were assembled.

7. Age and Churn

Age group	Customers	Exited	Churn rate
Under 30	1,641.0	124.0	7.56%
30–45	6,248.0	956.0	15.30%
46–60	1,647.0	842.0	51.12%
Above 60	464.0	115.0	24.78%



Source: calculations from European_Bank.csv (10,000 records, Year = 2025).

The most striking age-related pattern occurs among customers aged 46–60: 842 of 1,647 exited, giving a churn rate of 51.12%. Customers aged 30–45 contribute the largest number of exits (956) because they are the largest age group. A high rate and a high count are not interchangeable; both are useful for deciding where to investigate.

The under-30 group has a relatively low 7.56% churn rate. The above-60 group has 24.78% churn, but contains only 464 customers, so it contributes fewer exits than the 30–45 and 46–60 groups.

8. Geography × Age: Combined Risk

Country	Age	Customers	Exited	Rate
France	Under 30	866	45	5.20%
France	30–45	3168	376	11.87%
France	46–60	749	343	45.79%
France	Above 60	231	46	19.91%
Germany	Under 30	372	47	12.63%
Germany	30–45	1522	385	25.30%
Germany	46–60	502	338	67.33%
Germany	Above 60	113	44	38.94%
Spain	Under 30	403	32	7.94%
Spain	30–45	1558	195	12.52%
Spain	46–60	396	161	40.66%
Spain	Above 60	120	25	20.83%

The Germany × age 46–60 group has 502 customers and 338 exits, a **67.33%** churn rate. This segment alone accounts for 16.59% of all exits. The same age group also has elevated churn in France (45.79%) and Spain (40.66%).

Germany shows higher churn than France and Spain within every reported age band. This suggests that the German result is not explained solely by Germany having more middle-aged customers, although the data do not establish the remaining explanation.

9. Tenure and Credit Score

Tenure group	Customers	Exited	Churn rate
New (0–2)	2,496.0	528.0	21.15%
Mid-term (3–6)	3,977.0	821.0	20.64%
Long-term (7–10)	3,527.0	688.0	19.51%

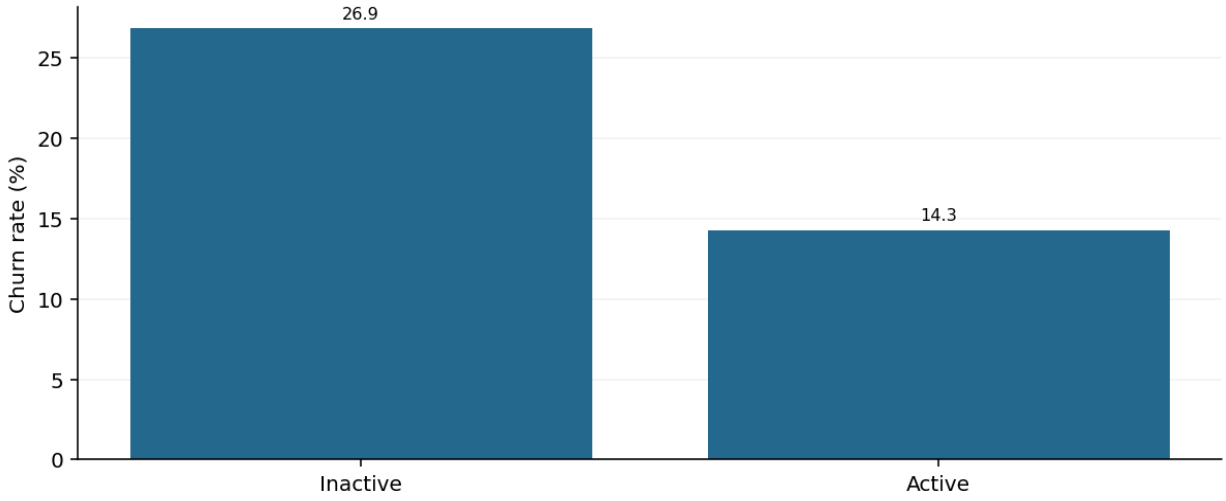
Churn varies only modestly by tenure: 21.15% for new customers, 20.64% for mid-term customers and 19.51% for long-term customers. Tenure is less discriminating in this dataset than age, geography or activity.

Credit score	Customers	Exited	Churn rate
Low	2,362.0	520.0	22.02%
Medium	4,490.0	893.0	19.89%
High	3,148.0	624.0	19.82%

Low-credit-score customers have a churn rate of 22.02%, compared with 19.89% for medium and 19.82% for high credit scores. The difference exists but is considerably smaller than the age and geography gaps. Very narrow credit-score bands can contain few customers; isolated extreme percentages should not be generalized.

10. Activity and Engagement

Activity	Customers	Exited	Churn rate
Inactive	4,849.0	1,302.0	26.85%
Active	5,151.0	735.0	14.27%



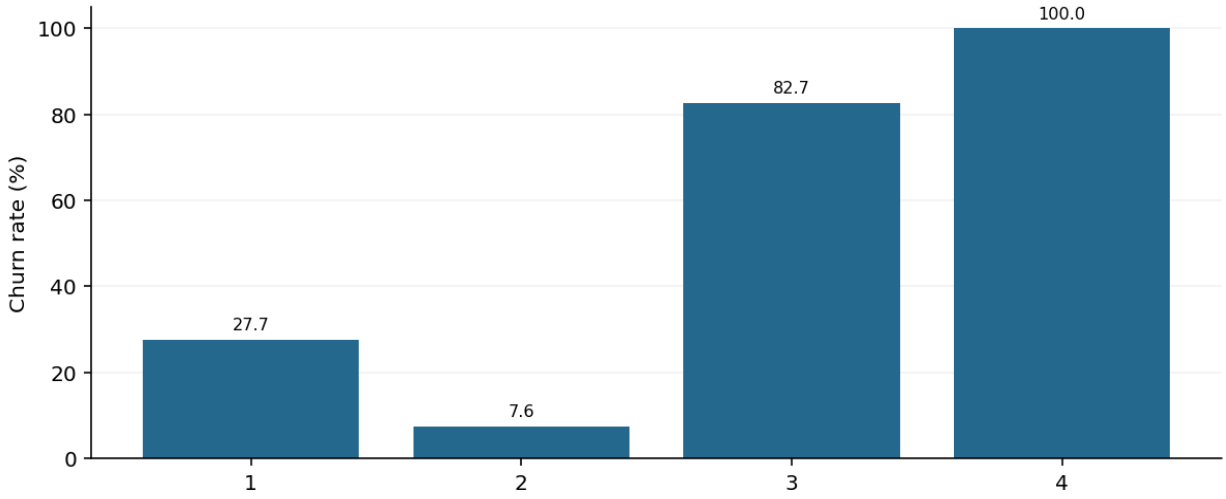
Source: calculations from European_Bank.csv (10,000 records, Year = 2025).

Inactive customers show a 26.85% churn rate, versus 14.27% among active customers. The inactive group accounts for 1,302 of 2,037 exits (63.92%). This is a strong descriptive association and a practical reason to examine customer engagement.

The highest geography × activity churn rate is found among inactive German customers: 41.08%. However, the data do not specify what qualifies as active or inactive, so a retention intervention should begin by validating that definition and investigating underlying behavior.

11. Product Ownership

Products	Customers	Exited	Churn rate
1	5,084.0	1,409.0	27.71%
2	4,590.0	348.0	7.58%
3	266.0	220.0	82.71%
4	60.0	60.0	100.00%



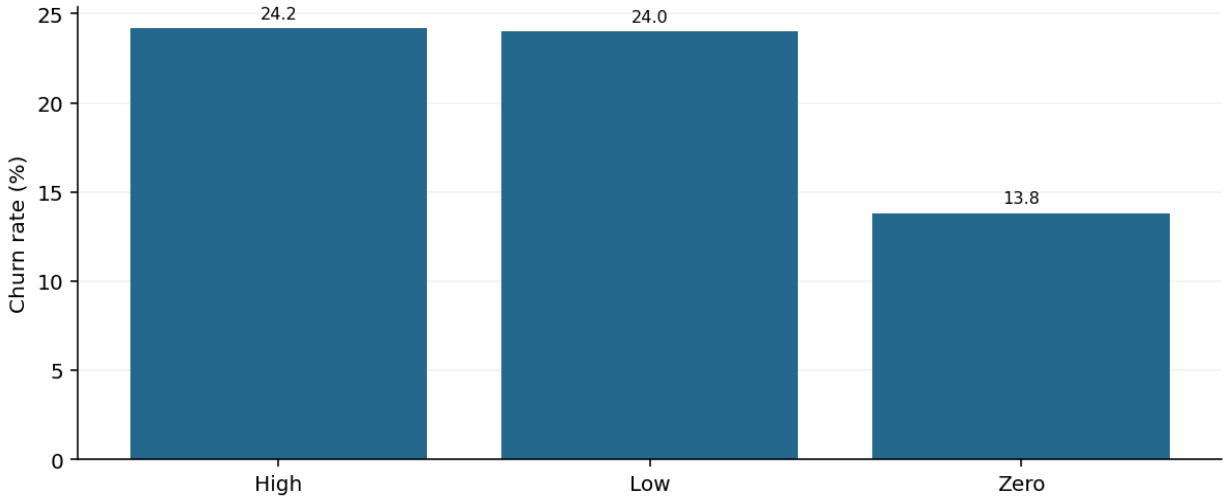
Source: calculations from European_Bank.csv (10,000 records, Year = 2025).

Customers holding two products have the lowest observed churn rate (7.58%). One-product customers have 27.71% churn and account for 69.17% of all exits. The three-product group has 82.71% churn, while all 60 customers holding four products are recorded as exited.

The unusually high rates among customers holding three or four products should be treated as a data-quality and research question. They may reflect selection effects, a small subgroup or features of the dataset itself. The results do not support a claim that adding products causes churn, or that cross-selling a second product would prevent exits.

12. Balance-Based Churn

Balance group	Customers	Exited	Churn rate
High	3,191.0	771.0	24.16%
Low	3,192.0	766.0	24.00%
Zero	3,617.0	500.0	13.82%



Source: calculations from European_Bank.csv (10,000 records, Year = 2025).

High- and low-positive-balance groups have similar churn rates: 24.16% and 24.00%. Zero-balance customers have a lower rate of 13.82%. The main observed separation is therefore between customers with a positive recorded balance and those with zero balance, rather than between the two positive-balance bands.

Germany has no zero-balance customers in the dataset. This matters when interpreting its country-level churn rate because zero-balance customers have relatively low churn in France and Spain. Nevertheless, Germany has higher churn even when comparing customers within positive-balance categories.

13. Financial Exposure: What the Balances Show

Total recorded account balances amount to **764,858,892.88** dataset currency units. Balances associated with exited customers total **185,588,094.63**, or **24.26%** of all recorded balances.

Category	Recorded balance
All customers	764,858,892.88
Exited customers	185,588,094.63
Retained customers	579,270,798.25
Exited balance share	24.26%

The phrase **churn-associated balance** is intentional. The dataset provides a balance snapshot and an exit flag, not a history of deposit movements. It cannot show how much money was withdrawn, whether accounts were transferred, or what the bank earned from these relationships. The totals are useful as an exposure indicator, not as a measured financial loss.

13.1 Exposure by country

Country	Churn-associated balance
France	57,666,164.54
Germany	97,973,915.53
Spain	29,948,014.56

Germany has the largest churn-associated balance at 97.97 million dataset units, followed by France at 57.67 million and Spain at 29.95 million. This reinforces the case for closer investigation of the German segment.

14. High-Balance Customers

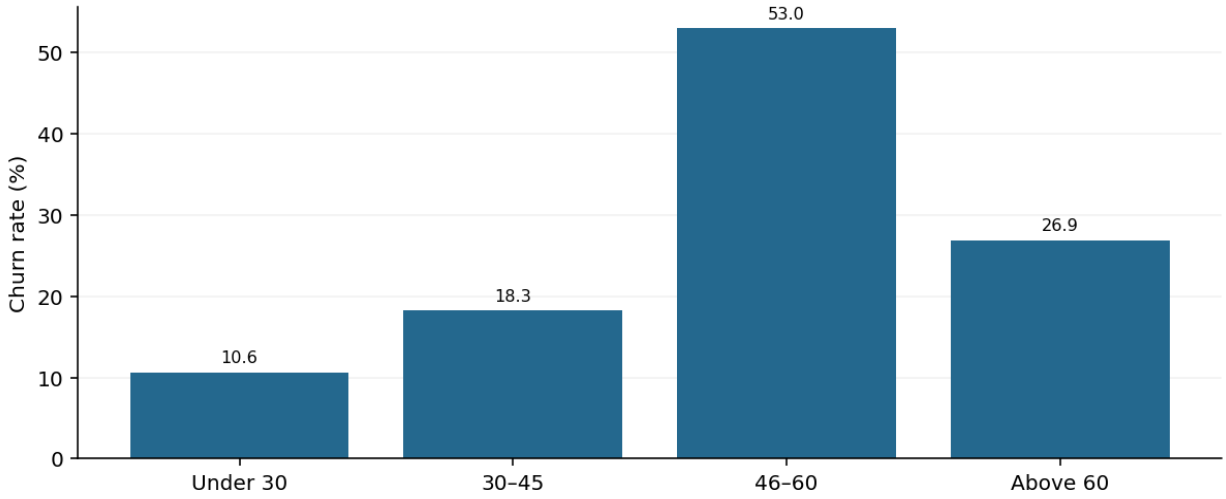
The high-value proxy selects the 2,000 customers with balances at or above **133,710.36**, the 80th percentile. Of these, 462 are recorded as exited, producing a **23.10%** churn rate. Their churn-associated balances total **71,676,672.51**, representing **38.62%** of all churn-associated balances.

Country	High-balance customers	Exited	Rate
France	861	179	20.79%
Germany	732	197	26.91%
Spain	407	86	21.13%

Germany has the highest churn rate among these high-balance customers (26.91%). Its churn-associated high-balance amount is approximately 28.83 million, only slightly above France’s 28.64 million. A decision based solely on country rank would overlook the very similar exposure in France.

15. High-Balance Customers by Age

Age group	Customers	Exited	Churn rate
Under 30	339	36	10.62%
30–45	1238	226	18.26%
46–60	330	175	53.03%
Above 60	93	25	26.88%



Source: calculations from European_Bank.csv (10,000 records, Year = 2025).

Among high-balance customers, those aged 46–60 have a 53.03% churn rate. Yet customers aged 30–45 contribute the largest high-balance churn-associated amount (35.05 million), because that age group is substantially larger. The 46–60 group contributes 27.21 million.

These findings illustrate the distinction between **risk intensity** and **aggregate exposure**. A smaller group can have the highest churn rate while a larger group contributes more total balance associated with exits. Management should consider both when setting research and retention priorities.

16. Main Findings and Discussion

Geographic concentration

Germany shows substantially higher churn than the other two countries. This pattern persists within age and balance segments, but its cause is unknown.

Middle-age concentration

Customers aged 46–60 have a 51.12% overall churn rate; the Germany × 46–60 segment reaches 67.33%.

Engagement association

Inactive customers have materially higher churn than active customers, suggesting a useful segment for investigation.

Product-count anomaly

Three- and four-product customers show unusually high churn, with very small samples for the four-product group. The pattern warrants validation before action.

Financial exposure

Churn-associated balances total 185.59 million units, and the top 20% by balance account for 38.62% of that amount.

Taken together, these results support a segmented retention approach rather than a single campaign for every customer. They do not establish that any specific customer attribute causes exit. Several variables overlap, and the descriptive comparisons have not been adjusted for those relationships.

17. Recommendations for Bank Management

Priority 1 — Investigate the Germany × age 46–60 segment

Review available service records, product terms and exit feedback for this group. Compare the segment with similar-age customers in France and Spain. The aim should be to identify plausible drivers before selecting an intervention.

Priority 2 — Test engagement-focused retention

For inactive customers, examine how activity is defined and whether low engagement precedes exit. A small, measurable outreach pilot could compare response and subsequent retention with a suitable control group.

Priority 3 — Review one-product relationships

One-product customers account for most exits and a large share of churn-associated balances. Investigate whether product suitability, service experience or customer needs explain the difference from two-product customers. Do not assume cross-selling is the remedy.

Priority 4 — Monitor high-balance exposure

Track churn rate and churn-associated balances separately for high-balance groups. Pay attention to both Germany and France, where high-balance exit exposure is similar. Avoid describing these balances as realized losses.

Priority 5 — Validate unusual product patterns

Audit definitions and record construction for the three- and four-product categories. The 100% churn figure for four-product customers is striking but based on only 60 records.

18. Dashboard and Reproducibility

The companion Streamlit dashboard is intended to let users filter customers by geography, age group, credit score group, tenure group, balance group, activity status, product count, gender and high-balance status. The dashboard calculates KPIs and visualizations from a prepared record-level dataset rather than relying on fixed summary figures.

A useful validation test is to confirm that the unfiltered dashboard reproduces 10,000 customers and 20.37% churn. Filtering Germany alone should produce 2,509 customers and 32.44% churn; adding age 46–60 should produce 502 customers and 67.33% churn. These are consistency checks against the analysis scripts, not independent proof that the original records are externally accurate.

Reproduction workflow

1. Load European_Bank.csv. 2. Validate record counts, types, missingness and identifier uniqueness. 3. Apply the documented segmentation rules. 4. Calculate grouped churn and balance summaries. 5. Export a record-level analytics file. 6. Launch the dashboard and compare key filtered outputs with the script results.

The dashboard should display a note that currency is unverified and churn-associated balances are not confirmed financial losses. Empty filter combinations should return an explanatory message rather than a misleading zero churn rate.

19. Limitations and Further Research

Dataset provenance

The dataset is provided for analysis without verified institutional provenance. It cannot support population-wide claims about European banks.

Cross-sectional design

All rows show Year = 2025. Without a longitudinal customer history or exit dates, the analysis cannot establish temporal sequences or trends.

Causality and confounding

Country, age, activity, products and balance can overlap. Segment-level differences are associations, not causal effects.

Financial interpretation

Recorded balances are not revenue, profit, withdrawn deposits or confirmed losses. Currency and balance timing are not documented.

Operational detail

No complaint, pricing, transaction, branch, service-quality or marketing-response fields are available.

Unusual patterns

Some product-count categories show extreme churn. These require scrutiny of subgroup size and source construction.

Next analytical steps

A later study could incorporate documented customer histories, service interactions and exit reasons; evaluate multivariable models with proper train/test separation; and assess retention interventions through controlled experiments. These are proposed extensions, not work claimed as completed here.

20. Conclusion

The study finds that churn is unevenly distributed across the supplied European banking customer records. Germany, customers aged 46–60, inactive customers and some product-ownership groups show comparatively high recorded exit rates. The Germany × age 46–60 segment stands out at 67.33% churn, while one-product customers account for the largest number of exits.

The financial analysis adds a second perspective. Churn-associated balances total 185.59 million dataset units, and the highest-balance fifth of customers account for 38.62% of that amount. This does not establish actual bank losses, but it identifies groups where further investigation could be financially relevant.

The most defensible management response is to validate the patterns, investigate likely reasons for exit and test targeted retention measures. The main contribution of this project is a reproducible, segment-level view of customer churn that distinguishes customer counts, churn rates and recorded balance exposure.

Data and analytical references

1. European_Bank.csv. User-provided dataset, 10,000 customer records, Year field = 2025. Source provenance and currency not specified.

2. Project Python analysis scripts (data validation, segmentation, churn summaries and high-value balance analysis), developed for this assignment.

3. Companion Streamlit dashboard, European Banking Intelligence (project implementation).

No external empirical sources are claimed in this report. A separate sourced literature review should be added if required by the internship assessment rubric.